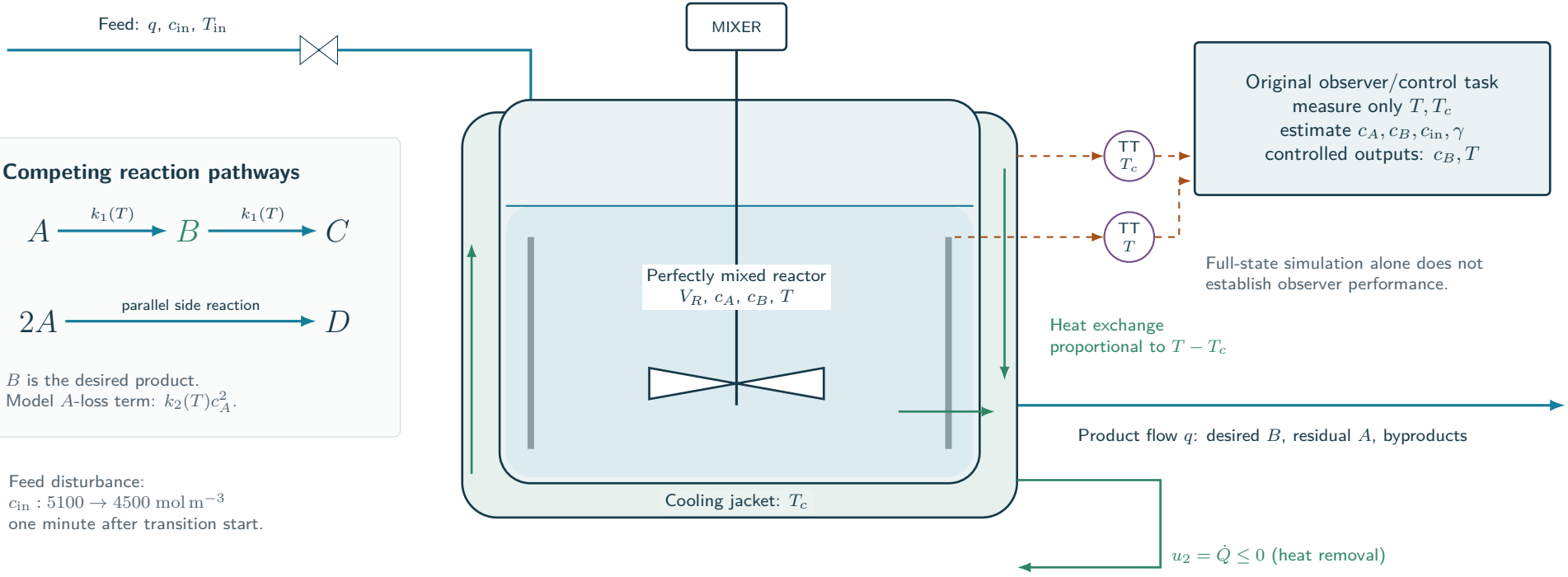
Klatt–Engell jacketed CSTR

Paper section 3.9: *Continuously stirred chemical reactor*

Four states: $[c_A, c_B, T, T_c]^\mathsf{T}$ | Inputs: dilution q/V_R and cooling capacity $\dot{Q}$ | Time: hours



Competing reaction pathways

$$A \xrightarrow{k_1(T)} B \xrightarrow{k_1(T)} C$$

$$2A \xrightarrow{\text{parallel side reaction}} D$$

B is the desired product.
Model A-loss term: $k_2(T)c_A^2$.

Feed disturbance:
$c_{in}: 5100 \rightarrow 4500\ \mathrm{mol\,m^{-3}}$
one minute after transition start.

Material balance and kinetics

$$\dot{c}_A = -k_1 c_A - k_2 c_A^2 + (c_{in} - c_A)u_1$$
$$\dot{c}_B = k_1(c_A - c_B) - c_B u_1$$
$$k_i = k_{i0}\exp[-E_i/(T + 273.15)].$$

Thermal balances

$$\dot{T} = h(c_A, c_B, T) + \alpha(T_c - T) + (T_{in} - T)u_1$$
$$\dot{T}_c = \beta(T - T_c) + \gamma\,(1000u_2).$$

h: reaction heat; 1000: MJ to kJ.

Units and input constraints

$5 \leq u_1 \leq 35\ \mathrm{h^{-1}}$; $u_1 = q/V_R$.
$-8.5 \leq u_2 \leq 0\ \mathrm{MJ\,h^{-1}}$.

c_A, c_B in $\mathrm{mol\,m^{-3}}$; T, T_c in °C. Kelvin is used inside Arrhenius rates.

Scope: Eqs. (39)–(43). Celsius resolves the Table 7 temperature-label inconsistency with Eq. (41). The jacket flow arrows illustrate heat-transfer hardware; the model input is cooling capacity, not a specified coolant flow.

Original explanatory drawing. Source: Kroll & Schulte (2014), DOI: 10.1016/j.asoc.2014.08.034. Model variants are identified in the accompanying case sheet.